

Problem 1 [40 points] Ensemble Learning: AdaBoost

Decision stumps are a popular family of binary classifiers used to implement the AdaBoost algorithm. Typically, they work as “weak classifiers”. A decision stump is a linear separator that is axis-aligned. In other words, it is a 1-level decision tree. It only has a root node. More formally, suppose the inputs are vectors in $\mathbb{R}_d$. A decision stump is a function h where

$$h(\mathbf{x}) = \begin{cases} 1, & \text{if } x_i \geq c \\ -1, & \text{otherwise} \end{cases}$$

for some $i \in d$ and $c \in \mathbb{R}$. (You may also change $\geq$ to $\leq$, the resulting function is still a decision stump.)

Point	x_1	x_2	y
1	1	3	-1
2	1	1	+1
3	2	2	+1
4	3	3	+1
5	3	1	-1

Figure 1: Dataset.

Suppose we have the data in Figure 1. On the first iteration of AdaBoost using decision stumps, we choose the classifier h where

$$h_1(\mathbf{x}) = \begin{cases} 1, & \text{if } x_1 \geq 1.5 \\ -1, & \text{otherwise} \end{cases}$$

1 [5 points]. What is the error ϵ_1 at the first iteration?

Error: $\epsilon_1 = 0.4$

2 [5 points]. What is the β_1 (the importance score of classifier h_1) at first iteration?

Importance score: $\beta_1 = \frac{1}{2} \ln\left(\frac{1-\epsilon_1}{\epsilon_1}\right) = 0.2027$

3 [10 points]. Which data points' weights will be increased?

Point 1 and Point 5

4 [10 points]. After updating in iteration 1, what are the weights of each data point (rounding to 4 decimal places)?

- Point 1: 0.1666
- Point 2: 0.2499
- Point 3: 0.1666
- Point 4: 0.1666
- Point 5: 0.2499

5 [10 points]. In next iteration, AdaBoost will train a new weak classifier $h_2(\mathbf{x})$ use the updated weights of data points. Which model in the follows would you choose as our classifier in the second iteration, why?

$$(A): h_2(\mathbf{x}) = \begin{cases} 1, & \text{if } x_2 \geq 1.5 \\ -1, & \text{otherwise} \end{cases}$$

$$(B): h_2(\mathbf{x}) = \begin{cases} 1, & \text{if } x_2 \leq 2.5 \\ -1, & \text{otherwise} \end{cases}$$

$$(C): h_2(\mathbf{x}) = \begin{cases} 1, & \text{if } x_1 \leq 2.5 \\ -1, & \text{otherwise} \end{cases}$$

$$(D): h_2(\mathbf{x}) = \begin{cases} 1, & \text{if } x_1 \geq 1.5 \\ -1, & \text{otherwise} \end{cases}$$

(C) is correct.

Problem 2 [60 points] Large Language Models

The **Midwestern Valley School District (MVSD)** serves roughly 18 000 students across 30 elementary and middle-school campuses. MVSD plans to pilot *EduAI Tutor*, a chat-based assistant that

- answers student questions about course material,
- explains concepts at multiple reading levels, and
- generates practice problems aligned with state standards.

Key Design Constraints

Requirement	Details
Knowledge scope	District-licensed digital textbooks (Grades 4–8, four subjects); teacher lesson plans; state test blueprints
Accuracy target	Factual error rate $\leq 3\%$ on sampled queries
Deployment	Runs inside the district’s private cloud (no outbound Internet)
Budget	\$25 k per year for compute & storage (commodity GPUs)
Privacy	Must not leak student PII or copyrighted content
Timeline	4-month proof-of-concept, then school-board review

Architectures Under Consideration

- 1. Baseline (In-Context Only)** Off-the-shelf 7-B parameter LLM (open weights).
 - Store textbook passages in a vector store and perform *retrieval-augmented generation (RAG)*.
 - “Stuff” retrieved passages + user query into the prompt; no parameter updates.
- 2. Enhanced (Fine-Tuned + RAG)** Same 7-B model, additionally fine-tuned on 60 k district Q&A pairs and 5 k teacher explanations.
 - Retains the retrieval pipeline but uses shorter prompts owing to domain adaptation.
 - Fine-tune cost: \$4 k one-time; refresh \$400 each semester.

Your task: Answer the following problems concisely:

1. Retrieval-Augmented Generation (RAG) [10 points]: The evaluation set has 120 student questions, each tagged with one or more gold passages. In an offline test, the retriever’s *top-5* list contained at least one gold passage for 87 questions. Compute Recall@5. If MVSD requires $\geq 80\%$ Recall@5 before launch, does the model pass?

$$\text{Recall@5} = \frac{87}{120} = 72.5\%$$

The current retriever fails to meet MVSD’s launch criterion.

2. Fine-Tuning vs. In-Context Learning [10 points]: List two technical *advantages* and two *disadvantages* of fine-tuning here, compared with pure in-context learning.

Advantage:

- Domain adaptation / shorter prompts: Fine-tuned models reduce prompt tokens, which leads to lower latency and serving cost.
- Higher faithfulness: Fine-tune can enforce citation style, reading-level adjustments, and task-specific formats, and lower hallucination.

Disadvantage:

- Maintenance overhead: Any curriculum change requires a new fine-tune cycle (data curation + compute).
- Risk of memorization / copyright leak: Additional parameter updates increase chance the model remembers licensed passages, triggering privacy or IP concerns.

3. Prompt Engineering [10 points]: Students sometimes type only “I don’t get fractions.” Propose a prompting strategy (few-shot, chain-of-thought, template switching, etc.) to elicit clarifying questions before answering.

- Clarification prompt (few-shot): “Could you tell me which part of fractions is confusing? 1. Adding, 2. Reducing, 3. Word problems, 4. Something else?”, which guides student to specify need; numbered choices speed response.
- Answer prompt (chain-of-thought concealed): “Think step-by-step but only show the final explanation”, which produces targeted explanation once clarification received; hides reasoning to reduce verbosity.

4. Hallucinations and Safety [15 points]: Define **hallucination** in LLM-based tutoring and give a concrete classroom example on history subject. Describe two complementary techniques: one *model-level*, one *post-generation* to detect or reduce hallucinations.

A **hallucination** occurs when the language model states or implies information that is not present in, or is contradicted by the retrieved curriculum sources. In a tutoring setting the error is especially harmful because students may treat the AI as an authority. For example, a 6-grade student asks, “Who invented the light bulb?” The model replies: “The light bulb was invented in 1872 by Nikola Tesla in New York City. Sources: Grade-6 Science Textbook, p.88.” The cited textbook page actually credits Thomas Edison (1879) and never mentions Tesla, so the answer both fabricates a fact and mis-attributes the source. It’s a classic hallucination.

- Model-level: After initial retrieval, run the generator twice with dropout seeds and compare factual claims.
- Post-generation: A separate scoring module (e.g., small transformer) scans the final answer to verify if every sentence has an in-scope citation.

5. Language-Model Selection [15 points]: MVSD could switch to a 13-B model instead of 7-B. Discuss trade-offs in *compute cost*, *latency*, and *knowledge capacity* for this K-12 use case.

- Compute cost: upgrading to 13-B may double the inference cost and could exceed MVSD’s annual compute budget, especially if utilization spikes.

- Latency: latency increase could negatively impact real-time responsiveness, especially during peak classroom hours.
- Knowledge capacity: The 13-B model's enhanced capacity may be beneficial for covering complex topics or nuanced phrasing.